Akash Balakrishnan

Los Angeles, CA | (408) 816 3138 | akashbal@usc.edu | linkedin.com/in/akash-balakrishnan

SUMMARY

Motivated MS in Computer Science student with hands-on experience in software development, cloud infrastructure, and debugging complex systems. Proficient in Python, C/C++, JavaScript, Node, distributed systems, virtualization, and machine learning operations. Actively seeking a Summer 2026 internship opportunity.

EDUCATION

University of Southern California(USC) **Aug 2025 - Aug 2027**

M.S. in Computer Science | Artificial Intelligence

- **Coursework:** Machine Learning, Analysis of Algorithms

University of California, Santa Cruz **Aug 2020 - Aug 2024**

B.S. in Computer Engineering (GPA: 3.7/4.0)

Coursework: Python Programming, Computer Systems & C Programming, Software Engineering, Database Management, Business Information Systems, Statistics & Probability, Object-Oriented Programming, Technical Writing, Computer Networks, Firmware Development, Logic Design, Computer Architecture, Parallel & Concurrent Programming, Signals & Systems, Electronic Circuits, Linear Algebra & Numerical Methods

TECHNICAL SKILLS:

Technical Skills: Python, C, C++, Java, JavaScript, SystemVerilog, Node.js, Flask, Django, Spiking Neural Networks (SNN), Prompt Engineering, Structured Data Extraction, AWS, REST API, MySQL, SQLite, FPGA (AMD Artix-7), Arduino, Raspberry Pi, BASYS3 Board, Vivado, PSViewer, Wireshark, Jira, Git, GitHub, Windows, UNIX, Linux, macOS, Agile/Scrum, Code Reviews

EXPERIENCE:

Polymorphics Robotics Lab (USC-ISI) **Sept 2024 - Present**

Graduate Researcher

- Design and implementation of mechanical tools and programs to diversify the capabilities of the self-reconfigurable robot Superbot

Oxford Instruments (OI: XT) **Jan 2024 - July 2024**

Software Engineering Intern

- Automated artifact detection on Beryllium disks using Python and C++, optimizing X-ray tube production to meet stringent buyer specifications
- Developed and programmed X-ray camera operations using local server commands and optimized them, which enabled precise power supply and pneumatic control
- Created artifact detection algorithms, improved accuracy, and reduced review time by 80%

UC Santa Cruz - SSN Word Identification **Jan 2024 - Mar 2024**

Undergraduate Research Assistant - Spiking Neural Network Word Identification

- Developed a speech-to-text system using Spiking Neural Networks (SNN) in Python.
- Collaborated on neuromorphic computing advancements, gaining expertise in backpropagation, activation functions, and hidden layers
- Implemented and debugged multithreading to enhance system performance

PROJECTS:

Anime AI **Jun 2023 - Jul 2023**

- Implemented a Flask and Node.js server for seamless backend functionality and developed APIs to integrate with the system
- Implemented a KNN-based recommendation system in Python to suggest anime titles based on user-selected genre preferences, with a RESTful API for interaction.
- Conducted efficient queries using an SQLite database populated through web scraping

Osmosys game development **Nov 2023 - Dec 2023**

- Integrated VGA-based Osmosys Game using BASYS3 Board using SystemVerilog.
- Proficient in FPGA development with Vivado, including FSM design, synthesis, timing analysis, testbench creation, simulation, bitstream generation, and validation
- Achieved precise VGA control, including Hsync, Vsync, and RGB signal generation, and successfully integrated a state machine for game flow, meeting all specified requirements

HTTP Server with Custom Firewall **May 2023 - Jun 2023**

- Designed multi-floor network with floor-specific subnets and switches
- Deployed HTTP server with custom firewall in simulated environment
- Implemented SDN router using C and C++ with POX Controller and custom protocol support

Huffman Encoder and Decoder **Mar 2023 - Apr 2023**

- Developed a robust Huffman encoder and decoder in C for file compression
- Incorporated Shannon entropy techniques to optimize compression by assigning shorter bit representations to frequently occurring symbols
- Employed static encoding for efficient compression with minimal data loss

RSA Encryption and Decryption in C **Feb 2023 - Mar 2023**

- Created a robust key generator, encryption, and decryption system using RSA algorithm in C

Small Company Network **Nov 2022 - Dec 2022**

- Designed and implemented a structured four-story network in Mininet using C and C++, configuring floor-specific switches, subnets, and dedicated server networks while optimizing IP routing with source/destination and port-based methods.
- Enhanced server security by blocking untrusted traffic and restricting ICMP, and developed POX controller rules for efficient network management with Wireshark-based performance analysis.

Hexapod Walker **May 2019 - Jul 2019**

- Designed and built an Arduino-based six-legged robot; developed control algorithms, assembled and soldered a custom PCB, and applied hardware engineering and soldering techniques to ensure precise and robust operation.
  https://youtu.be/EhPC9sjssrE https://youtu.be/2Ot8LESQMA4 https://youtu.be/Rc19IbLdJ6I

CERTIFICATIONS:

AWS Certified Cloud Practitioner **Sept 2025**
Advanced Learning Algorithms: DeepLearning.AI, Stanford University **Sept 2023 - Oct 2023**
Supervised Machine Learning: DeepLearning.AI, Stanford University **Sept 2023 - Oct 2023**